

Domácí úkol 1.1

Vypracovali: Daniel “*Localhost*” Ransdorf, Jan “*kokeshh*” Kodeš

Podpis:  

$$1. \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

$$\frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2) \stackrel{(*)}{=}$$

(*) : Skalární součin sama sebe je nezáporné reálné číslo, čili platí $(\sqrt{\langle \cdot \rangle})^2 = \langle \cdot \rangle$

$$\begin{aligned} &\stackrel{(*)}{=} \frac{1}{4} (\langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle - \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle) \\ &= \frac{1}{4} ((\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle) - (\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle)) \\ &= \frac{1}{4} (2\langle \vec{u}, \vec{v} \rangle + 2\langle \vec{v}, \vec{u} \rangle) = \frac{1}{2} (\langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle) = \frac{1}{2} (\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}) \\ &= \frac{1}{2} (\operatorname{Re} \langle \vec{u}, \vec{v} \rangle + i \cdot \operatorname{Im} \langle \vec{u}, \vec{v} \rangle + \operatorname{Re} \langle \vec{u}, \vec{v} \rangle - i \cdot \operatorname{Im} \langle \vec{u}, \vec{v} \rangle) \\ &= \frac{1}{2} (2\operatorname{Re} \langle \vec{u}, \vec{v} \rangle) = \underline{\underline{\operatorname{Re} \langle \vec{u}, \vec{v} \rangle}} \quad \square \end{aligned}$$

$$2. \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$

$$\begin{aligned} &\frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2) \stackrel{(*)}{=} \frac{1}{4} (\langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle - \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle) \\ &= \frac{1}{4} ((\langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, i\vec{v} \rangle - \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle) - (\langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle)) \\ &= \frac{1}{4} (-2\langle \vec{u}, i\vec{v} \rangle - 2\langle i\vec{v}, \vec{u} \rangle) = \frac{-1}{2} (\langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle) = \frac{-1}{2} (\langle \vec{u}, i\vec{v} \rangle + \overline{\langle \vec{u}, i\vec{v} \rangle}) \\ &= \frac{-1}{2} (i\langle \vec{u}, \vec{v} \rangle + i\overline{\langle \vec{u}, \vec{v} \rangle}) = \frac{-i}{2} (\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) \\ &= \frac{-i}{2} (\operatorname{Re} \langle \vec{u}, \vec{v} \rangle + i \cdot \operatorname{Im} \langle \vec{u}, \vec{v} \rangle - (\operatorname{Re} \langle \vec{u}, \vec{v} \rangle - i \cdot \operatorname{Im} \langle \vec{u}, \vec{v} \rangle)) \\ &= \frac{-i}{2} (2i \cdot \operatorname{Im} \langle \vec{u}, \vec{v} \rangle) = \underline{\underline{\operatorname{Im} \langle \vec{u}, \vec{v} \rangle}} \quad \square \end{aligned}$$